Design, Analysis, and Performance of a Silicon Photonic Traveling Wave Mach-Zehnder Modulator

David Patel



PREVIEW

Department of Electrical & Computer Engineering
McGill University
Montréal, Canada

December 2014

A thesis submitted to McGill University in partial fulfillment of the requirements for the degree of Master of Engineering.

© 2014 David Patel

Abstract

Silicon photonics is a technology platform for integrating passive and active optical devices. The ability to use existing CMOS-VLSI foundry processes to fabricate integrated optics enables large volume production at low costs while maintaining compatibility with electronic integrated circuits. With the recent explosion in network traffic, large datacenters are demanding higher and higher intra-datacenter transmission speeds over distances of up to 20 km. For this application, optical interconnects are more energy efficient and are able to satisfy the bandwidth demands of the foreseeable future. Optical modulators are an essential component of these communication links and immense research is dedicated to developing efficient high bitrate devices. It is desirable to have high bitrate transmission over a single channel because it simplifies multiplexing (e.g., fewer wavelengths and parallel fibers) which can increase costs. The work presented in this thesis uses the silicon photonic platform to realize an integrated Mach-Zehnder modulator (MZM) suitable for datacenter communication links.

A series push-pull traveling wave MZM is designed using T-shaped extensions to increase the microwave index. The modulator is designed to have a characteristic impedance close to 50 Ω for matching with common microwave drivers and terminations. In addition, the group velocities of the microwave and optical wave are closely matched to optimize the electro-optic bandwidth. A 37 GHz -3-dB electro-optic bandwidth was measured at 1 V reverse bias. The DC V_π was measured to be 7 V in the most efficient arm, corresponding to a $V_\pi L_\pi$ of 2.8 V-cm. For on-off keying modulation, open eye diagrams are visible up to 60 Gbps. Error-free operation with a bit error rate (BER) $<1\times10^{-12}$ was measured up to 45 Gbps, the limit of our photo-receiver. Eye diagrams with a clear opening were observed up to 70 Gbps using analog signal processing to compensate for the limitation of our bit pattern generator. The device was further characterized using digital-signal-processing to dispense the analog bandwidth limitations of the entire communication link to the digital domain. Using PAM-4 and PAM-8 modulation formats, 112 Gbps transmission on 2 km (5 km with PAM-8) of SMF fiber is achieved below the pre-forward-error-correction (FEC) BER threshold of 3.8×10^{-3}, corresponding to a post-FEC BER of $<1\times10^{-15}$. The best case energy consumed by the modulator is estimated to be 0.18 pJ/bit. The ability to generate 100 Gbps modulation in a single wavelength, as achieved in this work, is one of the main waypoints in realizing 400 Gbps and 1 Tbps short reach interconnects.

PREVIEW

Sommaire

La photonique sur silicium est une plate-forme technologique pour les dispositifs optiques intégrés passifs et actifs. La possibilité d'utiliser les procédés de fonderie CMOS-VLSI existants pour la fabrication de composantes d'optique intégrée permet une production à grande échelle et à moindre coût tout en maintenant la compatibilité des composantes avec les circuits électroniques intégrés. Avec les récentes explosions du trafic dans les réseaux, les grands centres de données exigent une vitesse de transmission interne de plus en plus grande sur des distances allant jusqu'à 20 km. Pour cette application, les interconnexions optiques sont plus efficace énergiquement et sont capables de satisfaire les demandes de bande passante à venir. Les modulateurs optiques sont des composantes essentielles de ces liens de communication et une grande partie de l'effort de recherche est dédiée à développer des dispositifs efficaces à haut débit. Il est souhaitable d'avoir des débits de transmission élevés sur un seul canal car cela simplifie le multiplexage (i.e. moins de canaux et de fibres en parallèle) qui pourrait augmenter les coûts. Le travail de recherche présenté dans cette thèse démontre l'utilisation de la photonique sur silicium pour réaliser un modulateur Mach-Zehnder (MZM) adéquat pour les liens de communication des centres de données.

Un modulateur Mach-Zehnder à ondes progressives, en série et en montage symétrique est conçu avec une extension en forme de T pour augmenter l'indice des micro-ondes. Le modulateur est conçu pour obtenir une impédance caractéristique de près de 50 Ω pour s'adapter aux impédances typiques de terminaisons et de pilotes à micro-ondes. De plus, les vitesses de groupe des ondes optiques et micro-ondes sont adaptés autant que possible afin d'optimiser la bande passante électro-optique. Une bande passante électro-optique de 37 GHz à -3 dB a été mesurée avec une tension à polarisation inverse de 1 V. La tension de polarisation V_π en courant continu mesurée était de 7 V, correspondant à un facteur de mérite $V_\pi L_\pi$ de 2.8 V-cm. Pour la modulation par tout ou rien (OOK), des diagrammes de l'oeil ouverts sont visibles jusqu'à 60 Gbps. Un fonctionnement exempt d'erreur avec un taux d'erreur sur les bits (BER) de moins de 1×10^{-12} a été mesuré jusqu'à 45 Gbps, la limite de notre photorécepteur. Des diagrammes de l'oeil avec une ouverture distincte ont été observés jusqu'à 70 Gbps en utilisant le traitement de signal analogue. Le dispositif a été analysé plus amplement en utilisant le traitement de signal digital afin de ne pas être limité par la bande passante analogue du lien de communication. En utilisant les formats de modulation PAM-4 et PAM-8, une transmission de 112 Gbps sur 2 km de fibre

PREVIEW

optique unimodale (SMF) a été réalisée sous le seuil de BER avec pré-correction d'erreur sans voie de retour (FEC) de $3,8\times10^{-3}$, correspondant à un post-FEC BER de moins de 1×10^{-15}. La consommation d'énergie du modulateur dans le meilleur des cas est estimée à 0,18 pJ/bit. La possibilité de produire une modulation à 100 Gbps avec une seule longueur d'onde, tel qu'accompli dans le travail présenté, est une des étapes importantes dans la réalisation d'interconnexion à courte portée à 400 Gbps et à 1 Tbps.

PREVIEW

Acknowledgments

I sincerely thank my supervisor, Prof. David V. Plant, for his constant encouragement in pursuing graduate studies. I am very grateful for the long discussions I had with him on various topics over the past few years, which he managed to organize despite his busy schedule. Moreover, I highly appreciate Prof. Plant's dedication to his work which has resulted in a laboratory with high-end equipment that is necessary to perform not only my research but also that of other students in the Photonic Systems Group at McGill.

I would also like to thank the current and past students in our group for their tremendous help and guidance: Venkatakrishnan Veerasubramanian, Samir Ghosh, Alireza Samani, Mathieu Chagnon, Mohamed Osman, Wei Shi, Imran Cheema, Zhaobing Tian, and Qiuhang Zhong. The experimental setup used for all silicon photonic related activity would be nonexistent without Samir and the contributions from Venkat, Ali, and Imran. I am particularly indebted to Samir, who had spent nearly as much time as I have in the laboratory to help me test my devices. Mathieu and Mohamed have contributed immensely to the DSP portion of this work. I acknowledge Wei Shi for initiating us to the topic of traveling wave Mach-Zehnder modulators. I thank Zhaobing Tian for introducing me to the workings of the PSG. I would like to reiterate my appreciation of Venkat and Samir for their advice on all issues in and out of the laboratory. I wish you all the best.

Furthermore, I would like to acknowledge Prof. Lukas Chrostowski from the University of British Columbia and NSERC for the CREATE Si-EPIC program. This program has taught me and numerous students nationwide the skills necessary to perform the design and layout of silicon photonic devices. In addition, I would like to thank the Canadian Microelectronics Corporation (CMC), in particular Dan Deptuck, who has made obtaining fabricated silicon photonic devices a reality. CMC also provided access to a wide suite of simulation and CAD tools that were indispensable for this work.

Moreover, I would like to thank our industry collaborators: Michel Poulin from TeraXion for his invaluable technical insight on modulators and Stephane Lessard from Ericsson Montreal for helping us obtain the DAC used in the experiment.

I dedicate this thesis to my parents and my brother. I would not be where I am if it was not for their hard work and perseverance. I am privileged of their unconditional support in my endeavors.

David Patel

PREVIEW

Contents

1 Introduction **1**
1.1 Silicon Photonics . . . 5
1.2 Silicon Optical Modulators . . . 7
1.3 Thesis Objective and Structure . . . 12

2 Silicon Mach-Zehnder Modulators **14**
2.1 Introduction . . . 14
2.2 Mach-Zehnder Modulator Basics . . . 14
2.2.1 Silicon Waveguide Design . . . 16
2.2.2 MZI Input/Output Coupling . . . 18
2.2.3 Unbalanced and Asymmetrical MZIs . . . 19
2.2.4 Silicon Thermo-optic Effect . . . 20
2.3 Transfer Matrix Model . . . 21
2.3.1 Modulator Metrics . . . 23
2.3.2 Extinction Ratio Dependence on Loss Imbalance . . . 26
2.3.3 Temperature Stability, Imbalance, and Fabrication Variations . . . 27
2.3.4 Modulation Chirp . . . 29
2.4 Modulator Driving Configurations . . . 30
2.4.1 Single and Dual Drive Modulators . . . 30
2.4.2 Series Push-Pull Modulator . . . 31
2.4.3 Power Estimates . . . 33
2.5 Modulation in Silicon . . . 35
2.6 p-n Junction Design . . . 39
2.6.1 Reverse and Forward Bias Diodes . . . 40

PREVIEW

2.6.2 Diode-waveguide structures . . . 42
2.6.3 Lateral p-n Junction Simulations . . . 43

3 Traveling-Wave Electrode Theory and Design 46
3.1 Introduction . . . 46
3.2 Distributed Analysis . . . 46
3.3 Coplanar Strips and Coplanar Waveguide . . . 48
3.3.1 Coplanar Waveguide . . . 49
3.3.2 Coplanar Strips . . . 50
3.3.3 Analytical Equations for CPW and CPS Transmission Lines . . . 50
3.3.4 Unloaded SOI Transmission Line . . . 53
3.4 Junction Loaded Transmission Line . . . 68
3.5 Electro-optic Small Signal Modulation Response . . . 72
3.5.1 Impedance Mismatch . . . 75
3.5.2 Microwave Loss . . . 77
3.5.3 Velocity Mismatch . . . 77
3.6 Segmented p-n Junction MZM Design . . . 79
3.7 Cascaded Matrix Model . . . 81
3.8 Bragg Frequency and Periodic Structures . . . 84

4 Device Design and Layout 85
4.1 Introduction . . . 85
4.2 p-n Junction . . . 85
4.3 Transmission line . . . 86
4.4 Modulator Length . . . 87
4.5 Electro-optic simulation . . . 88
4.6 Device Layout . . . 89
4.7 Testability . . . 90

5 Experimental Characterization and Discussions 91
5.1 Introduction . . . 91
5.2 DC and Optical Characterization . . . 91
5.3 On-chip Termination - DC and Small signal . . . 95
5.4 p-n Junction . . . 96

PREVIEW

5.5 Small Signal Characterization 97
5.6 Large Signal Characterization: OOK Modulation 100
5.7 Large Signal Characterization: OOK Modulation with ASP 103
5.8 Large Signal Characterization: OOK and PAM Modulation with DSP . . . 105
5.9 Modulator Energy Estimate 110

6 Conclusion **112**

References **115**

PREVIEW

List of Figures

1.1 Network traffic predictions for years 2013 to 2018 in zettabytes per year. The pie chart (inset) shows the predicted distribution of traffic in 2018. . . 2
1.2 An example of a short-reach optical transmit/receive chain with digital-signal-processing and error correction coding. 4

2.1 (a) Required change in effective index for a π phase shift as a function of the phase-shifting length for 1550 nm. (b) Effect on transmission spectrum due to a Δn_{eff} between two arms of a 3 mm long MZI. 15
2.2 Optical transverse electric (TE) mode at 1550 nm in silicon waveguides. . . 16
2.3 Simulated effective index in silicon waveguides 17
2.4 Simulated group index in silicon waveguides 18
2.5 Different configurations of MZI. 19
2.6 Change in effective index with temperature 20
2.7 Transmission of a MZM with applied voltage in (a) linear units, and (b) in decibel-milliwatts. 23
2.8 (a) Reduction in maximum achievable extinction ratio as a function of loss imbalance in the arms of a Mach-Zehnder interferometer. (b) Transmission spectrum of a single-drive forward bias modulator depicting the reduction in extinction ratio. 27
2.9 Simulated transmission characteristics of an ideal, lossless, 1×1 MZI. . . . 28
2.10 Dual-drive modulator driving configuration. 30
2.11 Dual-drive modulation with a source voltage V_S of 1 V_{pp} with both arms at the same bias voltage of -1 V. Voltage across each diode is shown in (a) and the resulting index change for both arms and with respect to the bias point is shown in (b). 31

PREVIEW

2.12 Series push-pull modulator driving configuration. 31
2.13 Series push-pull modulation with a source voltage V_S of 8 V_{pp} with a bias voltage of $-2\,V$. Voltage across each diode is shown in (a) and the resulting index change for both arms and with respect to the bias point is shown in (b). 32
2.14 Circuit models for series push-pull modulators in the (a) DC regime and (b) AC regime with the net voltages across the diodes listed in the figure. . . . 33
2.15 Change in attenuation, and refractive index in bulk silicon for different free carrier concentration. 38
2.16 Methods of plasma dispersion modulation in silicon: (a) carrier accumulation, (b) carrier injection, and (c) carrier depletion. 39
2.17 Small signal equivalent model of a p-n junction. 40
2.18 Examples of diode-waveguide structures for modulation. (a) Interdigitated (top view), (b) lateral p-n (top view), and (c) vertical (cross-section) junctions. 42
2.19 (a) Simulation flow for different lateral p-n junction geometries. (b) Lateral p-n junction with variable design parameters. 43
2.20 Simulated p-n junction (a) phase shift and (b) attenuation for a junction with a gap, and (c) phase shift and (d) attenuation for different offsets of the p-type semiconductor (negative axis distance indicates an offset in to the n region). All simulations shown are at 1.5 V reverse bias. 45

3.1 FEM simulation of (a) effective permittivity, and (b) critical length for a CPS electrode configuration. 48
3.2 Schematic of a (a) coplanar waveguide, and (b) coplanar strips transmission lines. 48
3.3 Analytical characteristic impedance of a coplanar strip transmission line for varying trace widths and spacing for the case of (a) 2 µm metal lines and an infinitely thick substrate, (b) infinitesimal metal line and an infinitely thick substrate, and (c) infinitesimal metal line and a finite 700 µm thick substrate. 53
3.4 Cross-sections of simulated CPS transmission lines. Three cases were considered: (a) CPS with last metal layer only, (b) CPS with entire metal stack including the first metal, last via, and silicon contact vias, and (c) CPS with entire metal stack of the same width as the top metal layer. 54

PREVIEW

3.5 FEM simulation of coplanar strips characteristic impedance at 30 GHz for three cases: (a) last metal only, (b) entire metal stack, and (c) entire metal stack of equal widths. 55
3.6 FEM simulation of coplanar strips attenuation at 30 GHz for three cases: (a) last metal only, (b) entire metal stack, and (c) entire metal stack of equal widths. 56
3.7 FEM simulation of coplanar strips effective index at 30 GHz for three cases: (a) last metal only, (b) entire metal stack, and (c) entire metal stack of equal widths. 56
3.8 Different CPS electrode designs: (a) narrow CPS, (b) slow-wave CPS, and (c) wider CPS transmission lines. 57
3.9 Simulated (a) microwave attenuation, (b) microwave index, and (c) characteristic impedance of CPS transmission lines 58
3.10 The calculated group and phase index of the slow-wave CPS design is shown in (a), and in (b) the absolute value of their difference is plotted as a function of frequency. 59
3.11 Equivalent RLCG model of a transmission line that is (a) unloaded, (b) loaded with a p-n junction, and (c) loaded with a p-n junction transformed into a parallel circuit. 60
3.12 Simulation of current volume distribution in the electrodes and vias at 30 GHz, (a) top view, and (b) side view. The current is denser at the edges that face each conductor. It is mostly concentrated on the edge of the bus electrodes and the bridge of the 'T' shaped extensions. 63
3.13 Simulated volume loss density (log scale) of the substrate and conductor at (a) 10 GHz and (b) 30 GHz. 66
3.14 RLGC parameters for the three CPS cases shown in Fig. 3.8. 67
3.15 FEM simulations of microwave loss for cases of transmission line without a p-n junction, a single p-n junction, and two p-n junction in series (for SPP). 71
3.16 Volume density loss in the metal and vias at 30 GHz, (a) top view, and (b) side view. 71
3.17 Volume density loss (logarithmic scale) in the metal, vias, and silicon at 30 GHz, (a) top view, and (b) side view. 72

PREVIEW

3.18 Graphical depiction of (a) small signal modulation response, and (b) electrical and optical bandwidth definitions. In (a), the amplitude of the input voltage (red) remains constant over a range of frequency while the corresponding modulated optical power (blue) decreases in amplitude for higher input frequencies, thus exhibiting a low-pass E-O modulation response. . . 73
3.19 Traveling wave electro-optic interaction. The net phase shift experienced by the optical wave is the integral of the change in effective index at each point of the transmission line. The $\Delta n_{eff}(V, z)$ depends on the attenuation of the microwave pulses (black) and group velocity mismatch with the optical pulses (orange) as they propagate along the transmission line. 74
3.20 Effect of characteristic impedance mismatch on the modulation response with perfect velocity mismatch and no microwave loss. 76
3.21 Effect of velocity mismatch on the optical -3 dB bandwidth, shown for (a) different index mismatch, and (b) for different lengths of the TWMZM. . . 78
3.22 Calculated requirements for the low-loss case of (a) junction capacitance, and (b) unloaded characteristic impedance for simultaneous matching of impedance and velocity. 80
3.23 Cascaded matrix modeling of TWMZMs. 82

4.1 Cross-section schematic of the SPP TWMZM design on an SOI wafer (drawing not to scale). 86
4.2 Layout schematic of the SPP TWMZM and a magnified view of the 'T'-shaped extensions (schematic not to scale). The dimensions are: $W = 120\ \mu m$, $S = 51\ \mu m$, $t = 2\ \mu m$, $S_T = 12.6\ \mu m$, $G_T = 9.2\ \mu m$, $W_T = 10\ \mu m$, $L_T = 47\ \mu m$, and $P_T = 50\ \mu m$. 87
4.3 Comparison of measured and simulated (a) E-E S_{21} response with R_{pn} and C_{pn} as fitting parameters, and (b) E-O S_{21} response using the electrical simulation of (a). Dashed lines mark -3 dB and/or -6.4 dB. 88
4.4 Experimental setup. 90

5.1 DC configurations for testing individual diodes of a SPP TWMZM: (a) diode 1 reverse biased, (b) diode 2 reverse biased, and (c) both diodes under the same reverse bias. 92

PREVIEW

5.2 I-V measurements for (a) forward bias, and (b) reverse bias diodes of the MZM. 92
5.3 Transmission spectrum measurements for (a) diode 1, (b) diode 2, and (c) both diodes under the same reverse bias voltage. 93
5.4 Phase shift calculated from transmission spectrum for (a) diode 1 and diode 2, and (b) both diodes under the same reverse bias voltage. 94
5.5 $V_\pi L_\pi$ calculated from transmission spectrum for (a) diode 1, and diode 2, and (b) both diodes under the same reverse bias voltage. 95
5.6 Measurements of an on-chip termination designed to 50 Ω. Saturation effect is observed in the (a) DC I-V measurement, and the (b) calculated DC resistance. Small signal measurements around different DC voltages is shown in (c). 96
5.7 Extracted p-n junction (a) capacitance, (b) resistance, and (c) the associated intrinsic bandwidth for different reverse bias voltages. 96
5.8 S-parameter measurements: (a) electrical to electrical S_{11}, (b) electrical to electrical S_{21} normalized to 1.5 GHz (dashed lines indicate -3 dB and -6.4 dB), and (c) electrical to optical S_{21} normalized to 1.5 GHz (solid line indicates -3 dB). 98
5.9 S-parameter measurements with and without the on-chip termination: (a) electrical to electrical S_{11}, and (b) electrical to optical S_{21} normalized to 1.5 GHz (solid line indicates -3 dB). 98
5.10 Extracted (a) attenuation, (b) microwave phase index, and (c) characteristic impedance from measured RF S-parameters. 99
5.11 Electro-optic response of this modulator and a shorter design without slow-wave electrodes. 100
5.12 OOK communication link. 100
5.13 Optical eye-diagrams of MZM at different bitrates with ER in dB and Q-factor in linear units. 102
5.14 Effect of drive and bias voltages for extinction ratio and receiver sensitivity. (a) Extinction ratio for diffferent drive and bias voltages. (b) BER measurement for received power at different bias voltages and a drive voltage of 4.8 V_{pp}. (c) Modulator energy consumption for error-free operation at 40 Gbps and the required optical power at receiver. 102

PREVIEW

5.16 Optical eye-diagrams of MZM at different bitrates with analog pre-emphasis. 104
5.17 DSP communication link. 105
5.18 Transmitter and receiver offline DSP. 106
5.19 Digital pre-emphasis filters applied at the transmitter for OOK, PAM-4, and PAM-8. 107
5.20 PAM-2 modulation results with DSP. Shown in (a) is the eye diagram, (b) BER for different bias voltages, and (c) BER for different transmission distances. 108
5.21 PAM-4 modulation results with DSP. Shown in (a) is the eye diagram, (b) BER for different bias voltages, and (c) BER for different transmission distances. 109
5.22 PAM-8 modulation results with DSP. Shown in (a) is the eye diagram, (b) BER for different transmission distances, and (c) the sensitivity of the receiver for the different modulation formats after 2 km of propagation. . . . 109

PREVIEW

List of Tables

1.1 Commercially available Mach-Zehnder modulators in other technologies compared with silicon Mach-Zehnder modulator from this work. 7
1.2 Recent developments of carrier-depletion based Mach-Zehnder Modulators 10

2.1 Comparison of different modulator driving configurations. 35

3.1 Analytical expressions for different types of CPW and CPS electrodes. . . . 53
3.2 Propagation constant and characteristic impedance of lines in different regimes. 62

5.1 Estimated power consumed by the SPP TWMZM. 111

PREVIEW

List of Acronyms

ASP	Analog-signal-processing
BER	Bit Error Rate
BERT	Bit Error Rate (Ratio) Tester
BW	Bandwidth
C-band	Conventional Band (1530 - 1565 nm)
CDR	Clock and Data Recovery
CPS	Coplanar Strips
DCA	Digital Communications Analyzer
DSP	Digital-signal-processing
EDFA	Erbium-doped fiber amplifier
E-O	Electro-optic
FEC	Forward Error Correction
ISI	Intersymbol Interference
LCA	Lightwave Component Analyzer
MIS	Metal Insulator Semiconductor
MZI	Mach-Zehnder Interferometer
MZM	Mach-Zehnder Modulator
NRZ	Non-return-to-zero
O-band	Original Band (1260 - 1360 nm)
OCT	On-chip Termination
OOK	On-off Keying
PAM	Pulse-amplitude Modulation
PD	Photo-diode
PR	Photo-receiver

PREVIEW

PRBS	Pseudorandom Binary Sequence
Q	Quality Factor
RF	Radio Frequency
RRC	Root-raised-cosine
SNR	Signal-to-noise Ratio
SOI	Silicon-on-Insulator
SPP	Series Push-Pull
SW-TWMZM	Slow-wave Traveling-wave Mach-Zehnder Modulator
TIA	Transimpedance Amplifier
TMM	Transfer Matrix Model
TPS	Thermal Phase Shifter
TWMZM	Traveling-wave Mach-Zehnder Modulator
VNA	Vector Network Analyzer
WDM	Wavelength Division Multiplexing

PREVIEW

"Pick a flower on Earth and you move the farthest star."

P. A. M. Dirac

PREVIEW

Chapter 1

Introduction

PREVIEW

The past decade has been marked by a significant increase in internet access media and an abundance of cloud-based storage, high-definition multimedia streaming, and services providing software, platforms, and infrastructures (SaaS, PaaS, and IaaS). This has resulted in an unprecedented demand in the speed and volume of data transfers. As forecasted by Cisco (Fig. 1.1), this demand will definitely continue to grow. By the year 2018, 8.6 zettabytes of data will be transferred per year [1]. What is astounding is that 75% of this network traffic is within the datacenter itself (east-west traffic pattern). On top of that, the internet of things or everything (IoT/IoE) is still nascent. It can be speculated that as IoT develops, it will further exacerbate network congestion within datacenters. Moreover, the massive volume of data has resulted in the creation of large ware-house scale datacenters. For such long and fast links, optical interconnects are believed to be superior to other forms of interconnects and are now becoming inevitable.

At the same time, the forefront and the future exascale supercomputers also require high bandwidth interconnects for transferring data from one central-processing-unit (CPU) to another, from the CPUs to the memory, and between different nodes in a cluster. Supercomputers consume large amounts of power, typically in the range of several megawatts, resulting in yearly energy costs of millions of dollars. Most of this power is consumed by interconnects. In fact, it is estimated that transferring data to the nearest chip can cost 200 times the energy needed to perform operations on that same data [2]. Thus, these interconnects must not only support high data rates but must also do this with very low energy consumption. In addition to supercomputers, better interconnects have also become

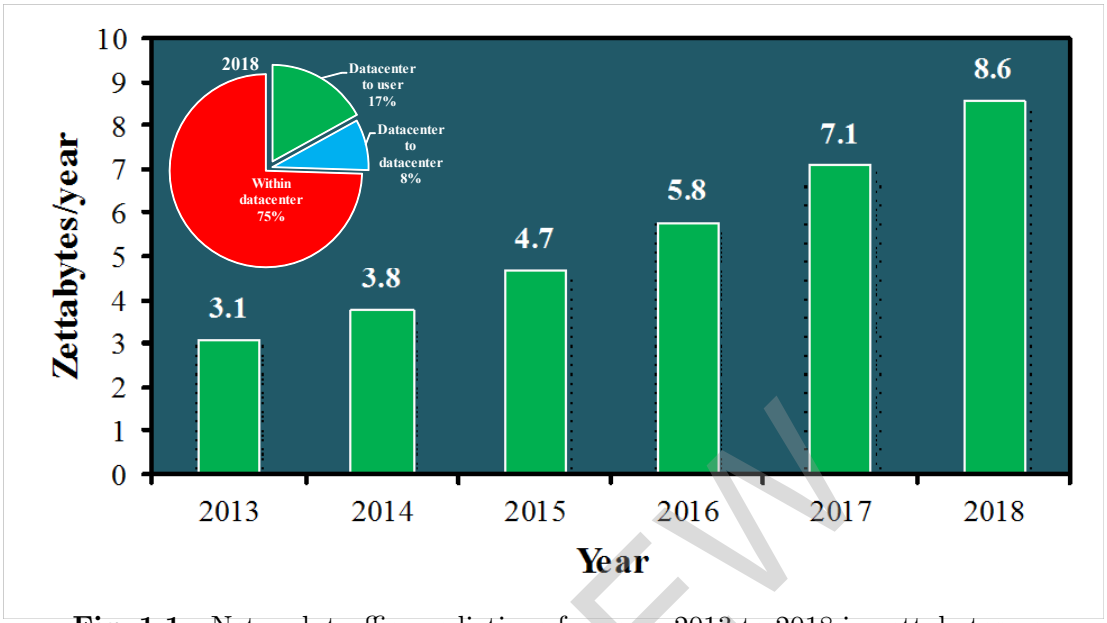


Fig. 1.1 Network traffic predictions for years 2013 to 2018 in zettabytes per year. The pie chart (inset) shows the predicted distribution of traffic in 2018. Figure produced from data in [1].

PREVIEW

necessary at a much smaller scale. The transit frequency f_T of transistors has reached hundreds of GHz with the recent CMOS technology nodes. Unfortunately, processor operating speed has remained significantly lower due to bandwidth limitation of the interconnect and the power consumption associated with them. High-speed on-chip clock distribution is also less reliable for higher frequencies with present electrical interconnects and is an additional obstacle to faster processor speeds.

Increased bandwidth demands at various scales has motivated tremendous research in the field of interconnects. Interconnects can be divided into two main categories: wireless and wireline. Wireless communication for chip-to-chip (inter-chip) and within the chip (intra-chip) is believed to be more convenient as complicated on-chip and on-board routing is avoided. It is also believed to be power efficient. However, the distance and the bandwidth is relatively limited. Wireless communication is also highly susceptible to interference, and protocols allowing only few devices to communicate at a time, or schemes using division multiple access need to be implemented [3]. The most common wireline interconnects are electrical and optical. Electrical interconnects have a long delay and are also very lossy. Therefore, they are only advantageous for low bitrates and short distances. Optical channels, on the other hand, have an extremely large bandwidth. For instance,

the conventional wavelength band (C-band: 1530-1565 nm) corresponds to a bandwidth of 4.3 THz. The optical channel can also be considered to be relatively lossless for short links. For example, a typical single-mode fiber (SMF) has a loss of only 0.2 dB/km in the C-band. Optical channels, compared to their electrical counterparts, have also very negligible cross-talk and thus high density interconnects are possible [4]. In fact, integrated optical paths can even cross each-other with negligible interactions among them. Although the optical channel may appear to be ideal, it does exhibit non-linearity and it is also quite dispersive in the C-band. These effects, however, become only an issue for longer distances and can often be compensated with signal processing.

While the optical channel is nearly ideal for short-reach communication, data needs to be generated and processed in the electrical domain. Compared to electrical interconnects, the additional steps of converting electrical signals to optical signals and vice-versa are required to make use of optical interconnects. This is a major hurdle in the adoption of optical interconnects as it increases cost and the total power consumption. Hence, for low bitrates and short distances, i.e., at the scale of intra-chip and inter-chip communications, copper interconnects cost less and are more energy efficient. The entire optical interconnect system would need to consume an energy of about 100 fJ/bit to be competitive [4]. The cost benefit begins to favor optical channels for longer distances and/or higher bitrates. For example, interconnects longer than 2-3 m at a bitrate of 25 Gbps favor optical channels over electrical interconnects in terms of cost. At 40 Gbps, optical interconnects are more cost effective than electrical interconnects for links longer than about 1 cm [2]. In a 16 petaflops supercomputer, optical interconnect would consume about 8.5% of the power consumed by a system with 10GBase-T cabling. This translates to savings of about $150 million dollars in operating costs over the life of the machine (∼10 years) [2]. Thus, for the applications of datacenter communication links and supercomputer nodes, optical interconnects are the most suitable choice.

Optical transmit-receive systems are typically implemented in a wavelength division multiplexing scheme (WDM), as illustrated in Fig. 1.2. The terahertz bandwidth of the optical channel is too large to be used adequately, therefore, it is usually divided into channels of smaller bandwidths with different center wavelengths. Typical wavelength spacing are 0.4 nm in dense WDM systems or 20 nm in a coarse WDM systems. Coarse systems can tolerate less accurate lasers as variations in the wavelengths can have less impact on neighboring channels, whereas dense systems allow for more channels within an

PREVIEW

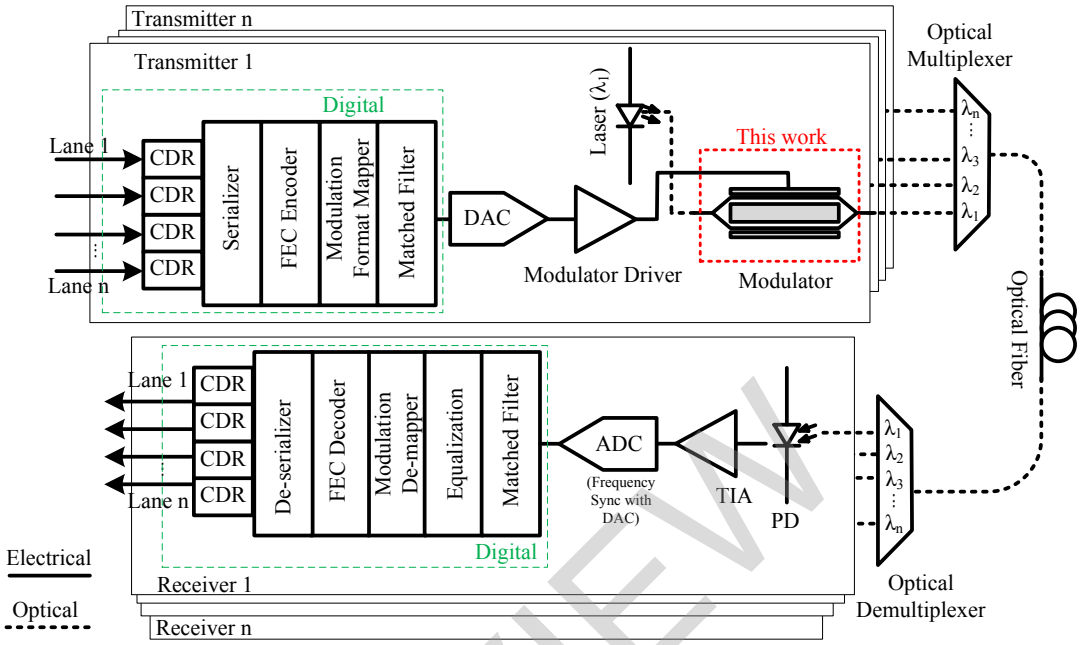


Fig. 1.2 An example of a short-reach optical transmit/receive chain with digital-signal-processing and error correction coding. CDR: Clock and data recovery, FEC: Forward error correction, DAC: Digital-to-analog converter, PD: Photo-diode, TIA: Trans-impedance amplifier, ADC: Analog-to-digital converter.

PREVIEW

aggregate bandwidth. The division of the large bandwidth into smaller bands can also reduce noise with bandpass filtering. Moreover, multiplexing allows the relatively slower electronics to make efficient use of the large optical bandwidth. The system in Fig. 1.2 also includes coding and digital-signal-processing (DSP) and would increase the total power consumed by the system. For short-reach communications, coding and DSP could only be justifiable when targeting extremely high datarates with very low bandwidth electronics and electro-optics. The data to be transmitted optically often comes into n electrical lanes operating at a slower bitrate B. A serializer will convert the parallel streams into a single stream at a bitrate of $B{\times}n$. Forward error correction coding makes it possible to recover data received with a high error count to one that is nearly error-free. This encoded data is then mapped to a desired modulation format, for example, multiple amplitude levels (such as PAM) in intensity modulated systems or quadrature amplitude levels in coherent systems. These modulation formats transmit symbols (multiple bits), instead of single bits within the same time slot and with about the same bandwidth. Thus, higher

spectral efficiency, which is defined as the ratio of the occupied bandwidth and the channel's available bandwidth, is achieved. The bit-to-symbol mapping is usually done with a labeling scheme that reduces the Hamming distance and optimizes the signal-to-noise ratio (SNR). After mapping to a modulation format, the data is pulse-shaped to reduce its bandwidth with a filter that is matched at the transmitter and the receiver. Using the same pulse shaping at the transmitter and receiver optimizes the SNR. The output of the digital-to-analog converter is then amplified to voltages required by the optical modulator. A laser provides continuous-wave light at the carrier frequency to be modulated. After propagating through the fiber, at the receiver side, the light is captured with a photo-receiver and digitized with an analog-to-digital converter (ADC). Operations in reverse of the ones at the transmitter are applied in order to recover the transmitted data. Clock and data recovery circuits are used to ensure that the continuous waveform is sampled at the appropriate time. In addition, the ADC and DAC should also sample with minimal frequency offsets between each other, though this can be corrected with additional digital-signal-processing (DSP).

This thesis presents the analysis, design, and testing of one of the sub-components of the optical transmitter: the Mach-Zehnder modulator (MZM) (outlined in red in Fig. 1.2). The MZM is essentially a transducer that converts data from the electrical domain to the optical domain and thus allows the use of the optical channel for data transmission. As such, it is a key component of transceivers. In the experiment discussed in this thesis, the MZM is operated in on-off keying (OOK) modulation and then with PAM modulation using DSP.

PREVIEW

1.1 Silicon Photonics

During the same time that bandwidth demands were increasing, significant development has occurred in the field of silicon photonics. Silicon photonic transmitters and receivers have even been commercialized. In this relatively new technology, optical integrated devices and circuits are being built on silicon, the material system employed by electronics for several decades. The silicon photonic fabrication process has matured over the past years, and many companies and institutes now have access to this technology. For instance, IBM has a 90 nm CMOS photonic process with monolithically integrated transistors, i.e., mixed signal (digital and analog) electronic circuits can be created on the same die as the optical circuits.

Luxtera also has a silicon photonic technology based on Freescale's 130 nm with monolithic integration. Other companies have acquired or also developed their own technologies, e.g., Intel, Cisco (acquired Lightwire), ST Microelectronics, Huawei (acquired Caliopa), and Mellanox. Some of these technologies have placed more emphasis on packaging and other forms of integration as oppose to monolithic integration of electronics, for example, hybrid integration techniques for lasers, and 3-D assembly technologies. Moreover, silicon photonic devices have also been built using unmodified IBM 45 nm SOI process with additional post-processing steps [5].

Most of these foundries have remained private, but various organizations worldwide have made multi-project wafer (MPW) runs available for fabrication of silicon photonic devices at foundries such as imec, CEA-LETI, VTT, IHP, and IME A*STAR. This has made possible for academia to obtain silicon photonic integrated circuits similar to microelectronics in the past. In Canada, this has been made possible by CMC Microsystems [6] and the NSERC CREATE Si-EPIC program [7]. The OpSIS-IME was a similar foundry service based in the United States.

The wide interest in silicon photonics arises from its numerous advantages. Since the technology uses existing CMOS-VLSI process steps, it is more reliable and has better yield compared to other integrated photonic technologies. This amounts to silicon photonics having an economy of scale resulting in cheaper integrated photonics for large volume production, such as the volumes expected for datacenter transceivers. Furthermore, it has the potential of monolithic, hybrid, and multi-chip module (system in a package) integration. For datacenter links, monolithic integration with electronics may not be optimal because optical devices are typically on the micrometer scale while electronics are on the nanometer scale. In advanced CMOS technology nodes, die area is expensive and the footprint of optical devices can significantly increase costs [8]. Furthermore, modifications required for monolithic integration with photonics can degrade transistor performance [8].

Another advantage of silicon photonics is its high index contrast ratio between the core and the cladding. This allows for denser optical devices and circuits relative to other technologies.

One of the biggest disadvantage of silicon is that the material has an indirect bandgap, thus it is very inefficient for light emission. Lasers will have to be integrated hybridly meaning additional fabrication steps and optical coupling losses. Furthermore, silicon is almost transparent at wavelengths of interest to telecommunications and photo-detectors

PREVIEW

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.